CE802-7-AU

UNIVERSITY OF ESSEX

Postgraduate Examinations 2016

MACHINE LEARNING AND DATA MINING

Time allowed: **TWO** hours

Candidates are permitted to bring into the examination room:
Casio FX-83GT PLUS or Casio FX-85GT PLUS only

The paper consists of **THREE** questions.

Candidates must answer **ALL** questions.

The questions are **NOT** of equal weight.

The percentages shown in brackets provide an indication of the proportion of the total marks for the **PAPER** that will be allocated for that part of the question.

Please do not leave your seat unless you are given permission by an invigilator.
Do not communicate in any way with any other candidate in the examination room.
Do not open the question paper until told to do so.
All answers must be written in the answer book(s) provided.
All rough work must be written in the answer book(s) provided. A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be marked.
At the end of the examination, remain seated until your answer book(s) have been collected and you have been told you may leave.

Candidates must answer ALL questions

<u>Question 1</u>

(a) Explain fully what is meant by the term *Markov Decision Process* (MDP) and explain the meaning of the elements of the tuple used to formally define a MDP. [10%]

(b) Explain the following equation in the context of *reinforcement learning*: [7%]

$$V^*(s) = \max_{a \in A(s)} [R(s,a) + \gamma V^*(s')]$$

(c) Consider a *reinforcement learning* problem modelled as a *MDP* with deterministic transitions and actions. The states are S = {A, B, C, D, E, G1, G2, G3} while the actions are trivially A = {toA, toB, toC, toD, toE, toG1, toG2, toG3}. The possible transitions and the corresponding rewards are indicated in the state transition diagram shown in Figure 1.1 below. Assuming a discount factor 0.5, calculate the discounted cumulative value of each state, also providing a brief explanation of the procedure followed. [13%]

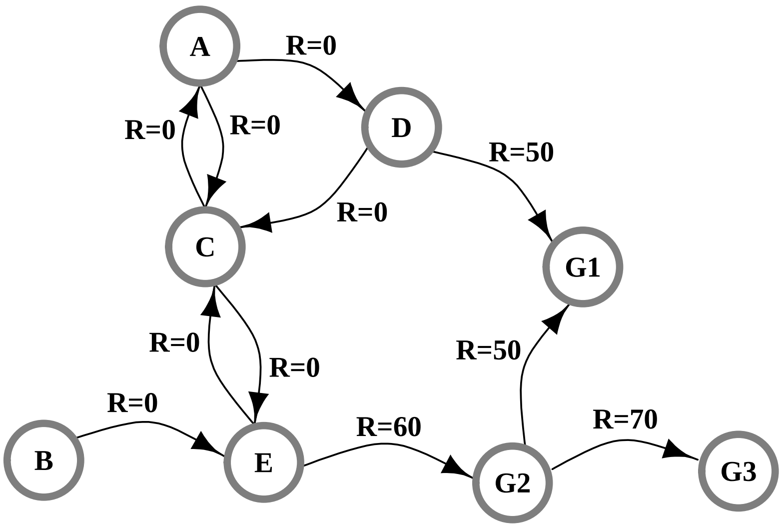

Figure 1.1

(d) Consider the application of Q-learning to a *reinforcement learning* problem modelled as a *MDP*. Assume there is a state s_0 with possible actions a_0, a_1, and a_2. After spending a long time learning, the agent has internal Q-values: $Q(s_0, a_0) = -6$, $Q(s_0, a_1) = -3$ and $Q(s_0, a_2) = -1$. What is the optimal action? Justify your answer. [5%]

Question 2

(a) Explain what is meant by the term *instance-based learning* and describe the essential components of a system that implements this learning technique. [10%]

(b) The simplest instance-based learning systems retain all training examples. Give *two* reasons why this is undesirable. [4%]

(c) Describe a successful method for selecting a smaller set of training examples for retention and explain, with the aid of a diagram, why it is effective. [11%]

(d) A marketing company uses survey data to identify prospective customers. They use a *k-nearest neighbour* system to learn to predict those people who are likely to buy the product. Among the attributes recorded in the survey are age, income, occupation and telephone number. Discuss what difficulties these attributes may cause for the learning system and suggest how such difficulties may be dealt with. [10%]

<u>Question 3</u>

(a) You are told that a given system achieves 95% accuracy on a binary (2 classes) classification task. What additional pieces of information do you need to know before you can decide whether this is a good result? List and discuss at least two. [8%]

(b) Testing a classifier's performance using a 10-fold cross-validation scheme produced the following confusion matrix: [10%]

		Predicted	
		Yes	**No**
Actual	**Yes**	39	11
	No	9	74

Table 3.1

Please indicate the values of TP, FP, TN, FN from the matrix, then compute the following measures:

(i) Estimated prior probability of class Yes

(ii) Recall (also known as true positive rate or sensitivity)

(iii) Precision (also called positive predictive value)

(iv) Classification accuracy

(c) Suppose that several different learning procedures have been applied to a classification task and that each has achieved partial success. Describe, in detail, a learning method that could be used to enable all the learning procedures to co-operate and make more accurate predictions. [12%]

END OF PAPER CE802-7-AU